Physiology of the Ovary and Uterus

Instructor: Katherine Lackritz (Katherine.Lackritz@jefferson.edu)

Block: Endocrine/ Urology/ Reproduction

Thread: Physiology

Conflict of Interest: Dr. Lackrtiz has no conflicts of interest to disclose

Learning Objectives

By exam time, you should be able to do the following:

1. Recall the Hypothalamic-Pituitary-Ovarian axis and how puberty is initiated
2. Describe the three phases of the menstrual cycle, including both ovarian (follicular) and uterine (endometrial) changes throughout the cycle
3. Describe the hormonal changes that happen throughout the menstrual cycle, focusing on LH, FSH, estradiol and progesterone
4. Explain the progression from primordial to dominant follicle and resultant ovulation and corpus luteum formation, including hormonal regulation
5. Summarize the process of meiosis in the oocyte and describe the process of fertilization.
6. Describe the physiology of sexual response and sexual dysfunction
7. Define peri-menopause and menopause and describe the most commonly experienced hypoestrogenic symptoms

Study Questions

1. What hormonal fluctuations occur during a typical ovulatory menstrual cycle (focusing on LH, FSH, progesterone and estrogen)?

2. What are the effects of estrogen and progesterone on the endometrium?

3. What occurs with the ovum between ovulation and implantation?

4. What are some types of sexual dysfunction and some of the factors that impact them?

Overview

This lecture is in order of the above LOs. We will review the initiation of puberty, then will talk about the menstrual cycle in detail, including uterine and ovarian cyclic changes, hormonal control of the cycle, and folliculogenesis, ovulation, fertilization, and (briefly) implantation. We will also review meiosis, which you have in more detail in your embryology modules from Dr. Spudich. We will then discuss sexual response and dysfunction in people with ovaries, and will end up with peri-menopause and menopause.

Recall the Hypothalamic-Pituitary-Ovarian Axis and How Puberty is Initiated

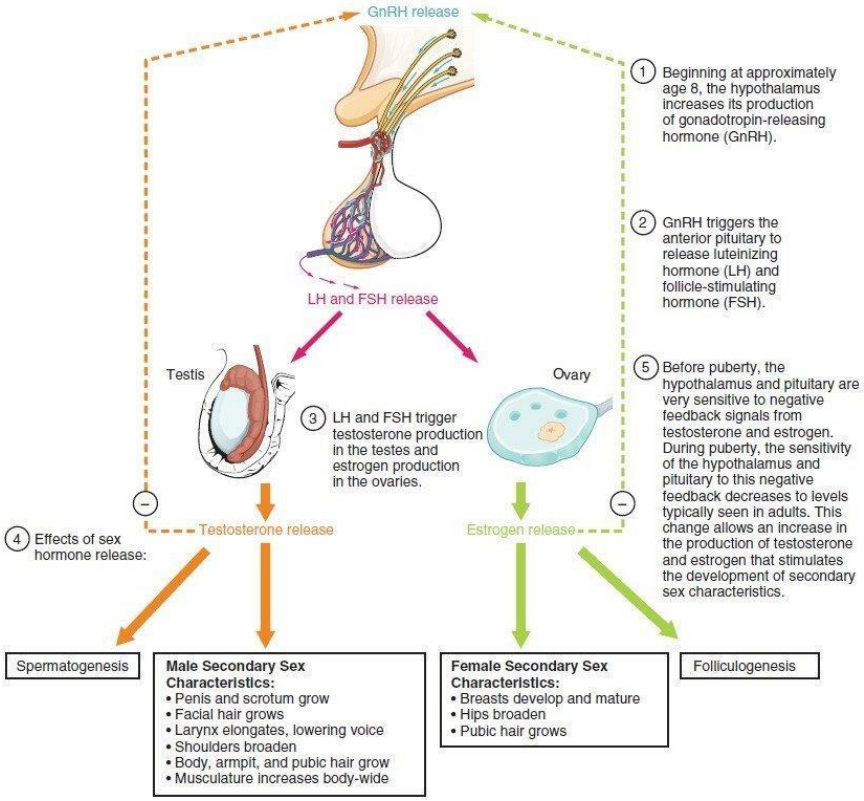
Maturation of the hypothalamic GnRH pulse generator initiates the onset of puberty. At midgestation LH and FSH levels are elevated to near adult levels. After birth the newborn is removed from placental and parental steroids, this stimulates gonadotropin secretion and there is a transient increase in testosterone for XY individuals and estradiol in XX individuals. These levels decline and then remain suppressed until the onset of puberty. Suppression happens through extreme sensitivity to negative feedback by steroids and CNS inhibition.

Puberty begins as the hypothalamic-pituitary axis becomes reactivated and inhibition wanes. The frequency and amplitude of the GnRH pulses increase. Initially the pulses happen at night, later they occur 24 hours a day.

There is a cascade of events that occurs from the GnRH pulses. Gonadotrophs are more sensitive and the secretion of LH and FSH is augmented, secretion of gonadal hormones is increased. These cause secondary sexual development to occur and establish negative feedback loops on the HPA axis. Later, LH and FSH secretion by the anterior pituitary and estradiol production in the ovaries leads to ovulation and menstrual cycles.

Physical signs of puberty are breast budding and the appearance of pubic hair. Axillary hair and the maximum growth velocity occur within one to two years.

Hormones DHEA and DHEAS are primarily responsible for the development of pubic and axillary hair. Adrenal maturation (adrenarche) precedes gonadal maturation by about 2 years. The pubertal growth spurt requires sex steroids and growth hormone to work together, GH is mediated by IGF-1.

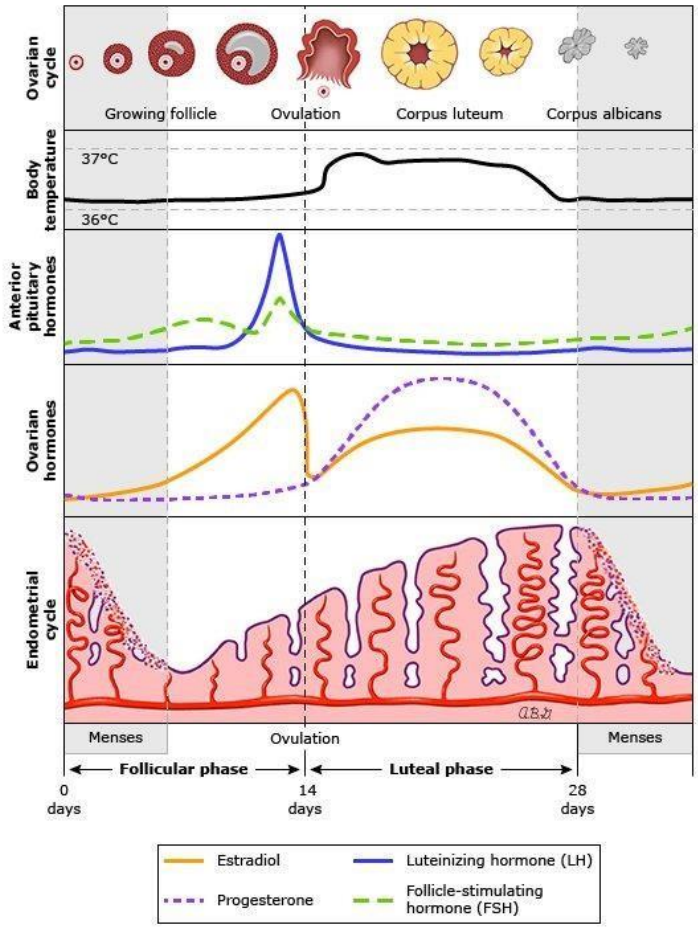


Describe the Three Phases of the Menstrual Cycle, Including Both Ovarian (follicular) and Uterine (endometrial) Changes Throughout the Cycle

Describe the Hormonal Changes that Happen Throughout the Menstrual Cycle, Focusing on LH, FSH, estradiol and progesterone

** Note: in the notes I have combined the information from these two LOs

The main physiologic goal is reproduction.



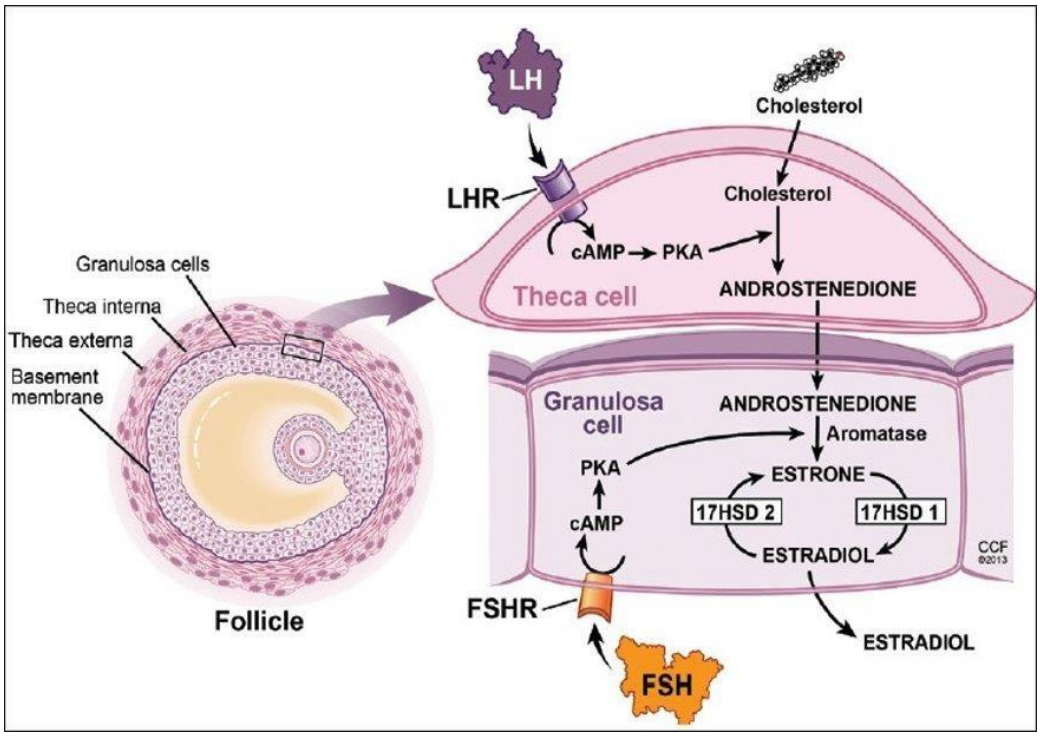
Follicular phase: This lasts from 10-16 days typically, from onset of menses to the day before ovulation. The cycle starts with low estradiol and progesterone levels. This leads to a decrease in negative feedback on the hypothalamus and an increase in GnRH pulse frequency and thus FSH and LH (though there are few LH receptors on the follicles at this time, so FSH is more important). In the ovary, we see FSH driven follicular development and selection of the dominant follicle, which will become the ovulatory follicle. The developing follicles make estradiol via theca and granulosa cells. Estradiol is the dominant ovarian hormone in the

follicular phase. Increasing estradiol eventually leads to a slight suppression of FSH and atresia of the non-dominant follicles. This happens because, at this time, the non-dominant follicles have fewer FSH receptors and less blood supply than the dominant follicle. The dominant follicle is stimulated by the remaining FSH and continues its growth. In the endometrium, at the start of the follicular phase we see shedding (menses, which starts after hormonal drop from previous cycle) followed by estrogen driven growth and thickening, as well as increase in glandularity. This is known as the endometrial "proliferative phase."

Mid-cycle surge and ovulation: Lasts about 24 hours. A switch occurs from estradiol being inhibitory to suddenly being stimulatory on the hypothalamus and pituitary. The resultant LH surge (and lesser FSH surge) occurs and leads to resumption of meiosis in the oocyte and ovulation of a mature oocyte (which is now a secondary oocyte), leaving behind the corpus luteum in the ovary. Estradiol must remain at a critical concentration prior to the LH surge and ovulation, so if it drops for any reason, ovulation will not occur that cycle. Premature estradiol drop may occur as oocytes (and the humans that house them) age and become less robust, leading to more anovulatory cycles as one approaches menopause. After the LH surge and ovulation, estradiol levels drop due to luteinization of the granulosa cells and final die off of nondominant follicles. Prior to the LH surge, progesterone must remain low, so a large bolus of progesterone prior to the LH surge will prevent that surge. This is how the emergency contraception "Plan B" works, by delaying or preventing ovulation. If a person were to take plan B after the LH surge, it would **not** work.

Luteal phase: Lasts about 14 days, from ovulation until the start of the next menses (the start of menses is the beginning of the next cycle). Ovarian hormones (estradiol and progesterone) again become inhibitory on the hypothalamus and pituitary. FSH and LH are suppressed by increasing estradiol and progesterone, which are made by the corpus luteum. Progesterone dominates in the luteal phase, where estradiol dominated in the follicular phase. The high amount of progesterone made by the corpus luteum causes cessation of mitoses (growth) in the endometrium and leads to stabilization, gland organization and spiral artery development in preparation for implantation of the embryo. The endometrium is now "secretory" and is producing fluid that is high in glycogen and early pregnancy regulatory molecules. These assist in implantation and early nutrition of the embryo (if it comes!). If no pregnancy occurs, the corpus luteum dies off and becomes the corpus albicans. The estradiol and progesterone (which were produced by the corpus luteum) drop and the endometrium destabilizes, becomes ischemic, and begins to shed, initiating the next cycle. As estradiol and

progesterone levels drop, GnRH increases leading to rise in FSH, initiating recruitment of the next cohort of follicles (for the following cycle).



Explain the Progression from Primordial Follicle to Dominant Follicle and Resultant Ovulation and Corpus Luteum Formation, Including Hormonal Regulation

Primordial follicle: these are formed in the human fetus before birth. People with ovaries are born with all of their oocytes that are capable of reproduction and do not make any more. This is different than sperm production, which happens throughout the lifetime of a person with testes. Primordial follicles are made up of a single primary oocyte surrounded by a single layer of flat granulosa cells and a basement membrane. The primary oocyte began meiosis 1 but arrested at Prophase 1 prior to birth. They have no independent blood supply and minimal endocrine influence, thus the progression from primordial to primary follicle is gonadotropin-**in**dependent. Over the years, many are lost to apoptosis, and some are "recruited" to be primary follicles, until very few/none are present after menopause. The number of primordial follicles remaining constitutes the "ovarian reserve"

Primary follicle: These are recruited from the primordial follicle reserve relatively constantly for the first 30 years of life. Various growth factors control recruitment, including many secreted by the oocytes themselves, not yet dependent on gonadotropins. In these follicles, granulosa cells become cuboidal but continue to be only a single layer. This is followed by enormous growth and differentiation of the oocyte. FSH receptors begin to be expressed on granulosa cells; this is necessary for further follicle development, which is gonadotropin-dependent. The zona pellucida (extracellular matrix) forms around oocyte. Its role is to protect the oocyte and to assist in appropriate fertilization by sperm. Intercellular connections between the oocyte and granulosa cells develop allowing regulatory molecule and nutrient passage. This allows communication within the follicle. The granulosa cells now change from simple cuboidal to stratified and proliferate significantly. There is a continued increase in gap junctions between granulosa cells for improved electrophysiologic communication. The theca cell layer forms outside the basement membrane. The theca cells are integral to ovarian steroid production later on. Angiogenesis occurs bringing circulation around the follicle, allowing nutrients, gonadotropins, and waste to pass back and forth. Overall, the secondary follicle contains a fully grown oocyte with zona pellucida, multiple layers of granulosa cells, a basement membrane, and the theca layer (or two) with capillary net. Meiosis continues to be inhibited; the oocyte is still stuck at Prophase 1.

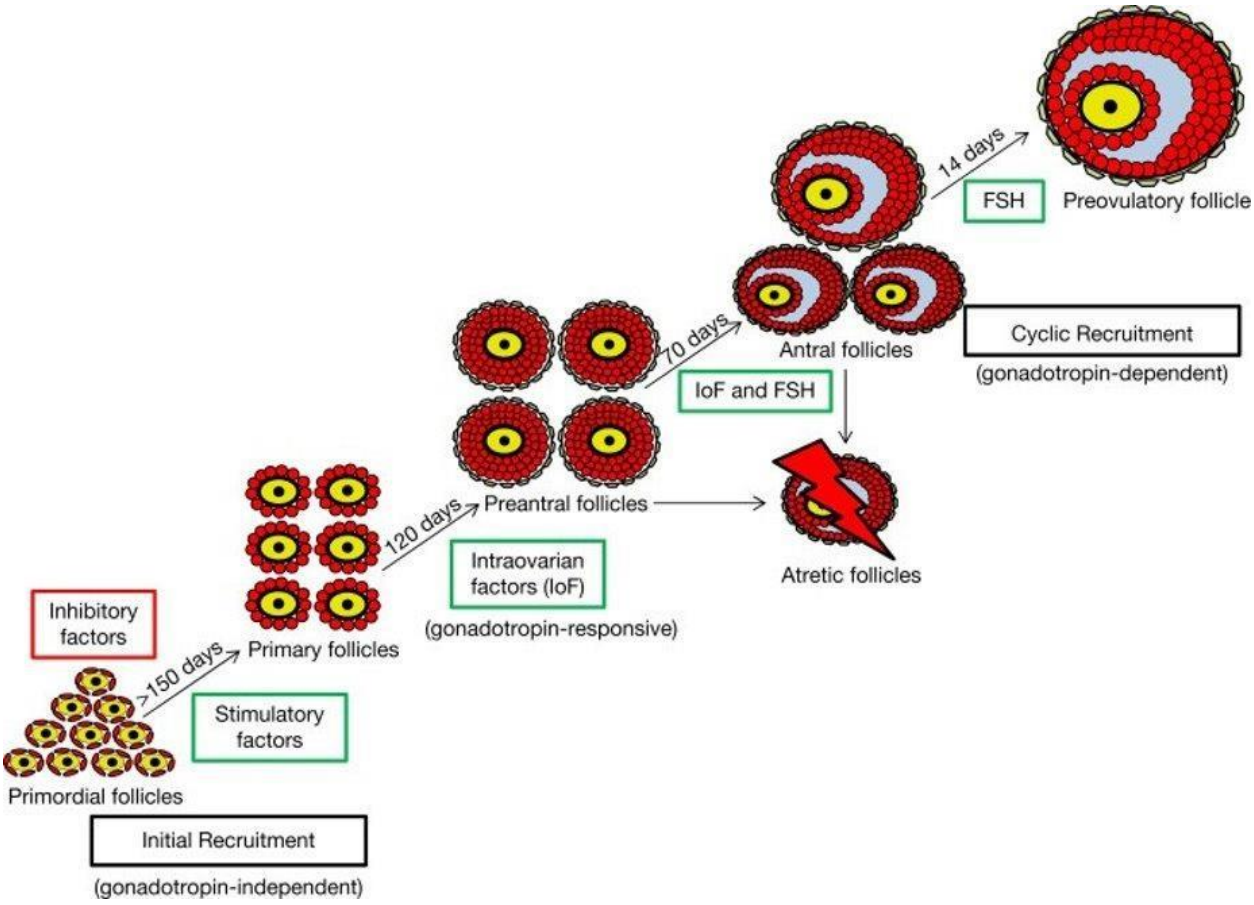
Secondary follicle (aka antral follicle): This is the first time we see follicular fluid, which makes this an “antral follicle.” The fluid is in the antrum, at one pole of the follicle, and is made up of plasma exudate from oocyte and granulosa cells. Regulatory molecules also pass through this fluid. The entire follicle can measure between 0.4mm (early on) to 25mm (about 0.98 in) in diameter and can be seen by

ultrasound. The follicle goes through enormous growth during this phase, but the structure remains largely the same. The theca cells divide into two layers, the theca externa, and the theca interna.

The "antral follicle count" is an ultrasound-based count of the number of antral follicles in the ovary and is used as part of fertility eval (proxy for ovarian reserve) and treatment planning.

Graafian follicle: Approximately 5-10 Graafian follicles begin to mature each menstrual cycle; one eventually becomes the dominant follicle, and the rest become atretic. Granulosa cells (FSH responsive) and theca cells (LH responsive) continue to proliferate, and the antrum continues to fill with fluid. The follicle that becomes dominant has up to 100-fold proliferation of granulosa and theca cells, and at least twice the vascular support compared to the non-dominant follicles, and higher density of FSH receptors. At ovulation, it measures 2.5 + cm and contains about 50 million granulosa cells and 7mL of follicular fluid! The theca externa differentiates into smooth muscle cells (which will help to expel the oocyte at the time of ovulation), and the theca interna cells become theca interstitial cells (steroid-producing) that have insulin and LH receptors and a rich capillary network. They produce high levels of androgens when stimulated by LH and insulin. This is part of the pathophysiology of PMOS, when patients have high insulin levels due to insulin resistance.

The dominant follicle has been moving through development from primordial follicle for about
300 days (10 months) prior to it being ovulated (see image below)



Typically only one follicle becomes dominant, though there are some people who have a genetic predisposition to polyovulation and are more likely to have fraternal (di-zygotic) twins. Identical (mono-zygotic) twins are a random event and not genetically inherited. Paternal family history of twins is also not relevant to the risk for twin gestations!

Over a lifetime, an estimate 400-500 follicles ovulate (estimate 40 years of menstrual cycles x 12 cycles per year = 480) and the rest undergo atresia. This number is decreased the more pregnancies one has (no ovulation during pregnancy and some of lactation), and if someone is on ovulation preventing contraception (more about this in your contraception lecture!)

Summarize the Process of Meiosis in the Oocyte

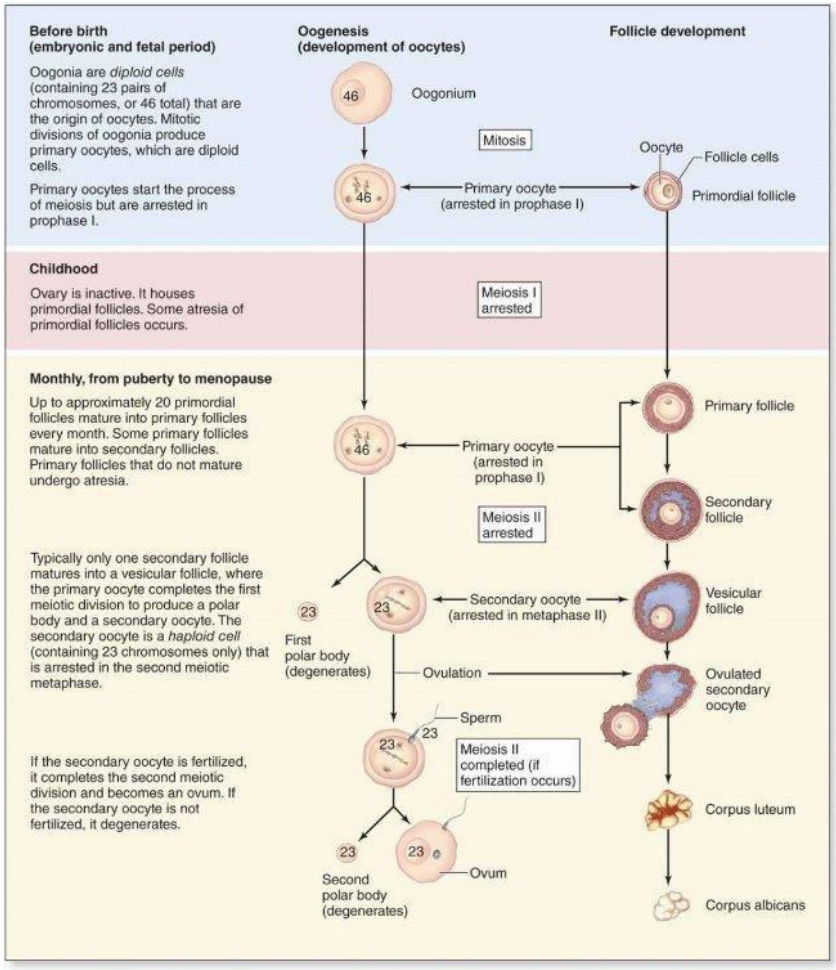
Please review Dr. Spudich's reproductive embryology module for more details on meiosis!

Meiosis involves DNA replication (during the S phase) followed by two meiotic divisions. The idea is to take one diploid cell and make 4 unique daughter cells

that are haploid (half the number of chromosomes as the parent cell). In the case of the oocyte, it makes two polar bodies and one ovum.

Primordial germ cells migrate to the fetal gonads and become oogonia. Oogonia undergo mitotic expansion then enter meiosis 1 to become primary oocytes. All healthy oocytes then arrest in Prophase 1 until ovulation when meiosis resumes, resulting in production of a mature gamete and the first polar body. The oocyte arrests again in metaphase 2, until fertilization and extrusion of the second polar body. After fertilization, it is a zygote (egg containing two haploid nuclei- from egg and sperm- which then combine into a single diploid nucleus)- any genetic issues here will pass on to all daughter cells. Mitotic divisions now grows cell number, developing an embryo then eventually a fetus.

The prolonged arrest in Meiosis 1 is what predisposes to mis-segregation of chromosomes and therefore aneuploidy. The older an egg is, the more likely its chromosomes are to separate incorrectly (non-disjunction) therefore increasing risk of aneuploidy (such as trisomy-21).

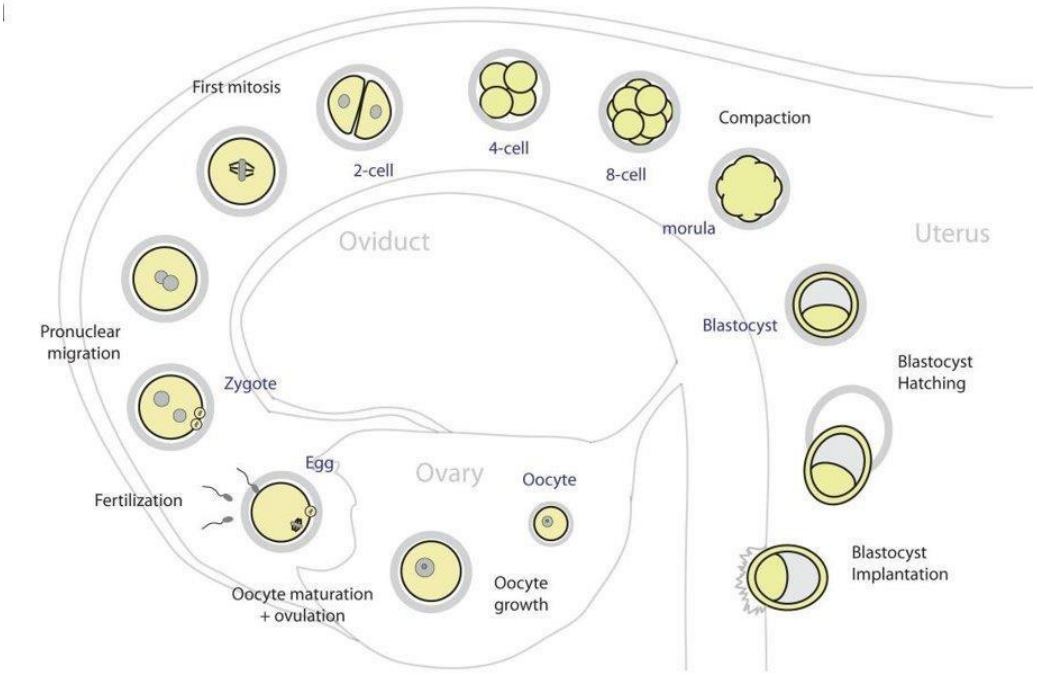


Describe the Process of Fertilization and Implantation

Sperm must be present in the tube at the time of ovulation. Leading up to ovulation, the uterus and cervix facilitate sperm transport through changes in cervical mucous and secretion from the endometrial glands. Estrogen thins cervical mucus to allow sperm to pass. Spinnbarkeit is the term for the capacity to stretch the cervical mucous without it breaking. Sperm undergo capacitation (removal of plasma coatings and surface molecules) allowing sperm to bind to the zona pellucida and penetrate the egg.

At the time of ovulation (as the oocyte leaves the ovary), meiosis 1 restarts in the oocyte. The egg then briefly pauses in metaphase 2 until the sperm meets the egg in the uterine (fallopian) tube. Fertilization occurs forming a zygote and the second polar body. The pro-nuclei (one from egg and one from sperm) then merge and mitosis and cell division occur as the zygote travels through the fallopian tube, first becoming a ball of cells called a morula, then a blastocyst, which ultimately implants into the uterine wall by day 9 or 10 after ovulation. At the time of implantation, the zona pellucida is shed and surface glycoproteins help the blastocyst adhere to endometrium. Junctional complexes form within endometrial cells and the blastocyst invades into the uterine wall. Trophoblast cells penetrate to initiate placental development. At this time, the body biochemically recognizes the pregnancy, blood supply is present, and pregnancy can be detected by testing human chorionic gonadotropin (hCG).

You will hear more about this during your upcoming fetal and placental physiology lecture!



Describe the Physiology of Sexual Response and Sexual Dysfunction

The human brain (limbic system) coordinates sex drive and mating. The sacral spinal cord is responsible for the in-the-moment physical changes that occur in the genitals.
Up to 25-45% of people with ovaries perceive they have sexual dysfunction.

There are two main sexual response models:

1. The linear model includes 1) Desire- flushing, increased genital blood flow/swelling followed by 2) Arousal- more swelling, tachycardia, muscle tension, 3)Orgasm- contraction of the vaginal muscles and finally 4) Resolution- return to normal functioning
2. The Intimacy based model recognizes these stages but adds emotional intimacy and emotional satisfaction, as well as biological, psychological, and psychosocial factors into the mix.

Many neurotransmitters/hormones have positive and negative effects on this cycle

- Positive: Norepinephrine, dopamine, oxytocin, serotonin (5-HT), testosterone, estradiol
- Negative: Prolactin, cortisol, GABA and serotonin via other receptors

Categories of sexual dysfunction include:

1. Hypoactive sexual desire: most common; lack of interest
2. Arousal disorder: desire present, but arousal difficult to achieve or maintain
3. Orgasmic disorder: difficulty achieving orgasm
4. Vaginal pain disorder (dyspareunia, vaginismus)

Remember: it is only a problem if it is a problem for the person or their

relationship Risk factors and causes of sexual dysfunction are as follows:

- Depression or anxiety
- Heart and blood vessel disease
- Neurological conditions, such as spinal cord injury or multiple sclerosis
- Gynecological conditions, such as vulvovaginal atrophy, infections or lichen sclerosus
- Certain medications, such as antidepressants or high blood pressure medications
- Emotional or psychological stress, especially with regard to your relationship
- A history of sexual abuse
- Low estrogenic states such as peri-menopause, menopause, postpartum, lactation, and premature ovarian insufficiency

Define Peri-menopause and Menopause and Describe the Most Commonly Experienced Hypoestrogenic Symptoms

Peri-menopause definition: Variable amount of time (can be years) before menopause when the number of healthy oocytes remaining is diminished and one or more of the following occurs:

- Menstrual cycle becomes irregular
- Estrogen level decreases
- Hypoestrogenic symptoms begin
- Progesterone decreases due to fewer ovulatory cycles
- Testosterone changes vary (mild decline to relatively stable)

Menopause definition: Cessation of menses for 12 months without other cause. During this time the following things occur:

- Few to no viable eggs left
- Very low estrogen and progesterone production
- Gradual decline in Testosterone from ovarian and adrenal sources
- Elevated FSH/LH levels

Common symptoms of hypoestrogenism include:

- Vasomotor symptoms: Hot flashes/night sweats
- Insomnia
- Brain fog/concentration difficulty
- Changes in mood and patience level
- Hair loss
- Decreased libido
- Genitourinary Syndrome of Menopause (usually later) o Urinary symptoms: urgency, stress incontinence o Vaginal dryness o Dyspareunia (pain with sex) Treatment options (briefly)

- Vasomotor symptoms: due to change in temperature sensation in the brain o HRT (estrogen or estrogen + progesterone) o Fezolinetant- *New* - Neurokinin B blocker, works in hypothalamus o SSRIs and SNRIs - increase serotonin levels, decreasing hyperthermia o Oxybutynin- anticholinergic used for OAB, blocks involuntary muscle activity o Gabapentin- antiepileptic, can also help with insomnia o Cognitive Behavioral Therapy
- Genitourinary syndrome: due to loss of estrogen effects on vagina and urethra- thinning of tissue and decreased self-lubrication o Topical estrogen therapy
- Mood changes
    - o SSRI/SNRI first choice as they also help with hot flashes

Signs of Reduced Progesterone post-menopause:

- Increased risk of endometrial hyperplasia (if unopposed estrogen)
- Sleep disturbances
- Mood variability

Signs of reduced Testosterone post-menopause:

- Low libido
- Reduced lean muscle mass
- Reduced bone density
- Fatigue
- Cognitive changes

Study Question Answers

1. What hormonal fluctuations occur during a typical ovulatory menstrual cycle (focusing on LH, FSH, progesterone and estrogen)?
   a. At the start of a cycle, during the menstrual/early follicular phase, the estrogen and progesterone levels are at their lowest. Low estrogen stimulates an increase in FSH which promotes follicular development. LH levels start low and remain low during this time. Estrogen levels then rise more rapidly due to increased production by the granulosa cells of the preovulatory follicle. When the level of estrogen is sufficiently high, it triggers a surge of LH, leading to final stages of meiosis in dominant oocyte and release of a mature gamete about 24 hours later. LH and FSH drop post-ovulation, as do estradiol levels. Progesterone increases after ovulation, produced by the corpus luteum. Estrogen levels rise again in the mid luteal phase, also made by the corpus luteum. The corpus luteum starts to involute and decreases production of estrogen and progesterone, starting the cycle again.

2. What are the effects of estrogen and progesterone on the endometrium?
   a. The quick drop in progesterone at the start of the cycle leads to breakdown and shedding of the superficial endometrium (stratus functionalis). The superficial endometrium then undergoes proliferation and thickening in response to rising estrogen levels. This is to prepare for implantation. After ovulation, progesterone stabilizes and maintains the stratus functionalis of the endometrium and causes it to become secretory. This is maintained until the corpus luteum dies off and the progesterone level drops, starting the cycle again.

3. What occurs with the ovum between ovulation and implantation?
   a. The oocyte is released from the ovary at the time of ovulation. It enters the Fallopian tube where, if sperm are present, fertilization occurs, and the second meiotic division occurs with emission of the final polar body. It is now a zygote, which travels through the Fallopian tube for 2-3 days during which time mitosis/cell division is occurring. The cell mass continues to grow and divide becoming a morula (spherical ball of cells) and then a blastocyst (ball of cells with fluid filled cavity). The blastocyst enters the uterus and implants into the endometrium. Implantation occurs around day 9-10 after ovulation.

4. What are some types of sexual dysfunction and some of the factors that impact them?
    a. Hypoactive sexual desire, arousal disorder, orgasmic disorder, and dyspareunia. They can be impacted by medications, medical conditions, and psychological factors. For example, SSRI medications used to treat depression or anxiety often result in decreased libido. Dyspareunia secondary to vaginal dryness may result from low estrogen. Depression and body image concerns, as well as relationship and life demands can also impact sexual function. It is often multi factorial.